Sportano is a famous brand that makes sports equipment such as boots, jerseys, footballs, and more. Four different sizes of balls designed for different age groups are made in three manufacturing sites. Each site has labor and machine-hour limitations, and the values for each are given in Tables 1–3. The process of manufacturing the balls in all production sites is similar, but not quite identical. Hence, each production site uses different amounts of raw material in creating the same ball size. The material requirements for each football are given in the last column of Tables 1–3. The amount of raw material available for the entire production during the planning period is 3,500 pounds.

Sportano has 3 major customers for its products: Royal Spanish Football Federation (RFEF), The Football Association (TheFA), and The French Football Federation (FFF). The maximum demand for each customer is given in Table 4 for each product. Tables 5-7 show the selling price of each ball size, the transportation costs, and the manufacturing costs of each product for each production site.

Sites 1 and 2 are experiencing a problem with production that is causing an unusual rate of defects in the balls. Based on that, all shipments manufactured in sites 1 and 2 that are used to fulfill the demand for the customers RFEF or TheFA must go through a special inspection to ensure that the balls conform to the standards. This is done by sending the manufactured balls to a central inspection site where they get tested, and then they get sent to their final destination. The capacity of this inspection site is 1,500 balls during one planning period.

Answer the following questions:

a. Determine Sportano's optimum plan for the production and demand fulfillment.
b. Find the total cost and revenue associated with the total production for each manufacturing site.
c. Marketing is trying to convince RFEF to purchase 50% more footballs in a period. Can the current resources in hand accommodate this increase or do we need to add more resources? How much more money can we make if we take on the additional demand?

Table 1: Product-Resource Constraints: Site 1

	Resources		
Football Type	**Labor (Hours/Unit)**	**Machine (Hours/Unit)**	**Material (Lb./Unit)**
Size 2	3	8	1.0
Size 3	3	8.5	1.1
Size 4	4	9	1.2
Size 5	4	9	1.3
Total available	6000	10,000	---

Table 2: Product-Resource Constraints: Site 2

	Resources		
Football Type	**Labor (Hours/Unit)**	**Machine (Hours/Unit)**	**Material (Lb./Unit)**
Size 2	3.5	7	1.1
Size 3	3.5	7	1.0
Size 4	4.5	8	1.1
Size 5	4.5	9	1.4
Total available	5000	12,500	---

Table 3: Product-Resource Constraints: Site 3

	Resources		
Football Type	**Labor (Hours/Unit)**	**Machine (Hours/Unit)**	**Material (Lb./Unit)**
Size 2	3	7.5	1.1
Size 3	3.5	7.5	1.1
Size 4	4	8.5	1.3
Size 5	4.5	8.5	1.3
Total available	3000	6000	---

Table 4: Maximum Product Sales

k

		Customers		
	Football Type	**RFEF**	**TheFA**	**FFF**
	Size 2	200	400	200
i	**Size 3**	300	300	400
	Size 4	500	200	300
	Size 5	200	400	300

j

i

Football Type	Labor (L_{ij})		
	Site 1	Site 2	Site 3
Size 2	3	3.5	3
Size 3	3	3.5	3.5
Size 4	4	4.5	4
Size 5	4	4.5	4.5

j

i

Football Type	Machine (M_{ij})		
	Site 1	Site 2	Site 3
Size 2	8	7	7.5
Size 3	8.5	7	7.5
Size 4	9	8	8.5
Size 5	9	9	8.5

j

i

Football Type	Raw Material (R_{ij})		
	Site 1	Site 2	Site 3
Size 2	1.0	1.1	1.1
Size 3	1.1	1.0	1.1
Size 4	1.2	1.1	1.3
Size 5	1.3	1.4	1.3

Table 5: Product Sales Price ($) per Unit

k

i

Football Type	Customers		
	RFEF	**TheFA**	**FFF**
Size 2	17	16	16
Size 3	18	18	17
Size 4	22	22	23
Size 5	29	26	27

Table 6: Shipping Costs ($) per Unit

k

j

Production Site	Customers		
	RFEF	**TheFA**	**FFF**
1	1.0	1.6	1.1
2	1.2	1.5	1.0
3	1.4	1.5	1.3

Table 7: Production Costs ($) per Unit

j

i

Football Type	Production Site		
	1	**2**	**3**
Size 2	14	13	14
Size 3	16	17	15
Size 4	18	20	19
Size 5	26	24	23

Decision Variables

x_{ijk} = the number of footballs of size i produced at manufacturing site j *which will go to customer k*, where i = 2, 3, 4, 5, j = 1, 2, 3, and k = 1, 2, 3

~~y_j = choosing manufacturing site or not (1 or 0)~~

Objective Function

Maximize the total ~~revenue~~ (profit), which is the sum of revenue from selling each type of football to each customer minus the total production cost and transportation cost.

Formulate Constraints

Demand fulfillment constraint: The total number of footballs of each size sold to each customer must not exceed the maximum demand.

Resource constraints: The labor, machine, and material resources used in each manufacturing site must not exceed their respective limits.

Inspection site capacity constraint: The total number of balls from sites 1 and 2 sent for inspection must not exceed the capacity of the inspection site.

Non-negativity constraint: The number of footballs produced must be non-negative.

P_{ik} = the selling price of a football of size i to a customer k ~~at manufacturing site j~~.

C_{ij} = the production cost of a football of size i at manufacturing site j.

T_{jk} = the transportation cost ~~of a football of size i~~ from manufacturing site j to a customer k,

D_{ik} = the maximum demand of a football of size i to a customer k.

L_{ij}, M_{ij}, R_{ij} = the labor, machine, and material requirements for producing a football of size i at manufacturing site j

L_j^{max}, M_j^{max}, ~~R_j^{max}~~ **(R^{max}=3500 pounds)**, I^{max} = the maximum labor, machine, and material resources available at manufacturing site j and the maximum capacity of the inspection site, respectively.

Objective Function

$$Maximize\ Z = \sum_{i=2}^{5}\sum_{j=1}^{3}\sum_{k=1}^{3}[(P_{ik} - C_{ij} - T_{jk}) \times x_{ijk}]$$

Subject to:

Demand

$$\sum_{j=1}^{3} x_{ijk} \le D_{ik}\ \ for\ i = 2,3,4,5\ , and\ k = 1,2,3$$

Resource

$$\sum_{i=2}^{5}\sum_{k=1}^{3} L_{ij} \times x_{ijk} \le L_j^{max}\ \ for\ j = 1,2,3$$

$$\sum_{i=2}^{5}\sum_{k=1}^{3} M_{ij} \times x_{ijk} \le M_j^{max}\ \ for\ j = 1,2,3$$

$$\sum_{i=2}^{5}\sum_{j=1}^{3}\sum_{k=1}^{3} R_{ij} \times x_{ijk} \le R^{max}\ \ (where\ R^{max} = 3500\ pounds)$$

Inspection

$$\sum_{i=2}^{5}\sum_{j=1}^{2}\sum_{k=1}^{2} x_{ijk} \le I^{max}$$

Non-negativity

$$x_{ijk} \ge 0\ \ for\ all\ i, j, k$$